Arm A2: Forward-Risk P-OPD

A first-principles tutorial on supervised evidence, Bayes calibration, and trust projection

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Abstract

Arm A2 is a proposed ODE-compatible distillation rule. An old student first produces a behavior endpoint. That endpoint is then re-noised along an independent forward flow-matching bridge, on which a frozen teacher and frozen old student compete to predict the sampled velocity label. Their squared-error difference becomes a detached sigmoid gate, while a separate metric projection bounds the teacher displacement used as a regression target.

This tutorial derives the gate from a shared-covariance predictive-residual model, proves the conditional risk decomposition and local replay gradient, and makes the event-sum, event-mean, temperature, resolution, and trust-radius scales explicit. It also distinguishes the exact auxiliary likelihood from practical calibration, A2 from transition-mixture P-OPD (A3), and forward-probe training from the ODE solver that generated the behavior endpoint.

Status boundary

Arm A2 is a theory proposal, not a claim about an existing implementation. The existing target mode named `xopd_target_mode: p_opd` denotes **Arm A3**, the stochastic transition-mixture construction. Arm A4 is implemented under `xopd_target_mode: marginal_cfm`; the existing `direct` mode is only A0-like and is not an exact implementation label for the A0/A1 taxonomy used here. Nothing in this tutorial asserts that an A2 API, configuration field, helper, trainer branch, or logging key already exists.

Contents

1 The probability object comes first

Fix a condition $C = c$. Every distribution and expectation below is conditional on C, even when C is suppressed to keep notation readable. A2 uses three layers, and only one of them supplies its evidence:

1. an ODE *behavior law* produces x_{data};
2. a forward FM *supervised coupling* produces (z, u);
3. an auxiliary *predictive-residual likelihood* interprets the gate.

It does not compare state densities and does not mix ODE marginals or stochastic transitions.

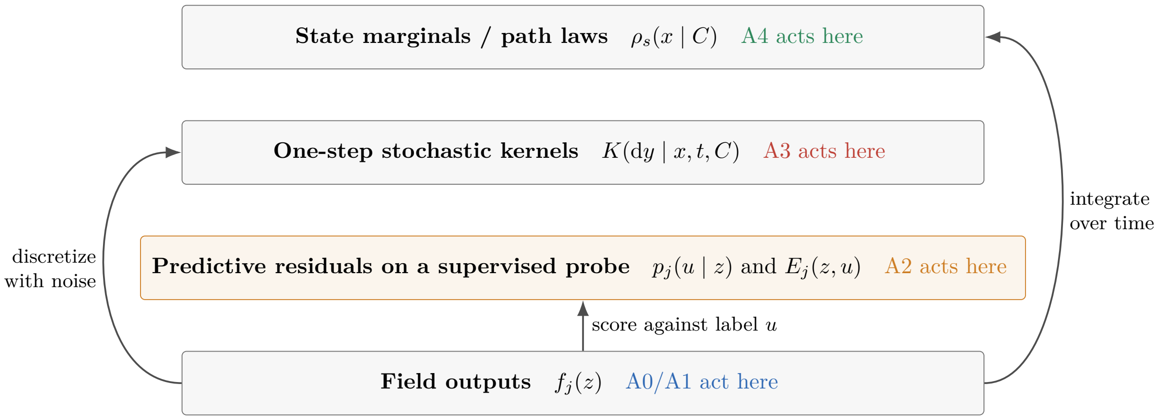


Figure 1: Probability-object layers. A2 evaluates predictive risk on a sampled FM label. Moving its formulas to the transition or marginal layer changes the method.

> **One-sentence definition**
>
> A2 asks whether the frozen teacher predicts a fresh forward-flow target better than the frozen old student on an old-generated endpoint, then moves a detached regression target toward the teacher by a separately trust-bounded amount.

2 From behavior rollout to a forward FM probe

2.1 Behavior endpoint

Let the old student define the ODE used for behavior generation. In a noise-to-data time coordinate t,

$$\frac{\mathrm{d}X_t}{\mathrm{d}t} = v_{\mathrm{old}}(X_t, t, C), \qquad X_0 \sim \rho_{\mathrm{base}}(\cdot \mid C), \qquad x_{\mathrm{data}} := X_1^{\mathrm{old}}. \tag{1}$$

The ODE solver may use any numerical grid appropriate to the model. Once it returns x_{data}, A2 discards the solver path for the purpose of constructing its supervised probe.

2.2 Fresh forward coupling

Independently draw

$$s \sim \omega_{\mathrm{FM}} \text{ on } [0, 1], \qquad \varepsilon \sim \mathcal{N}(0, I_D), \tag{2}$$

and define

$$x_s = (1 - s)x_{\mathrm{data}} + s\varepsilon = x_{\mathrm{data}} + su, \qquad u = \varepsilon - x_{\mathrm{data}}, \qquad z = (x_s, s, C). \tag{3}$$

This bridge has *data-to-noise* orientation: $s = 0$ is the old-generated endpoint and $s = 1$ is fresh Gaussian noise. Its derivative is exactly $\partial_s x_s = u$. A generation scheduler may run in the opposite direction; the distinction is a coordinate/sign convention, not a reason to reuse rollout timesteps. The fresh s and ε decouple the training probe from the ODE solver steps. This is the standard linear conditional flow-matching coupling [1], specialized to old-generated endpoints as formalized in the internal five-arm note [4].

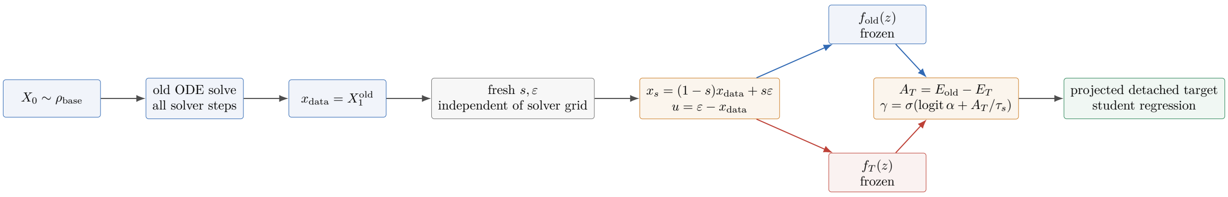


Figure 2: Full A2 pipeline. The behavior ODE supplies only the endpoint. The FM probe uses fresh randomness, and gradients belong only to the final student evaluation.

Listing 1: Good conceptual probe construction; names are illustrative, not APIs.

```
# A conceptual A2 iteration.
x_data = solve_old_behavior_ode(initial_noise, condition).detach()
s = sample_fm_time(batch_size)
epsilon = sample_standard_normal_like(x_data)

x_s = (1.0 - s) * x_data + s * epsilon
u = epsilon - x_data

with frozen_evaluation():
    f_old = old_model(x_s, s, condition)
    f_teacher = teacher_model(x_s, s, condition)
```

Listing 2: Bad: coupling the probe to stored ODE steps changes the evidence distribution.

```
# WRONG for A2: rollout states and solver labels are not the fresh FM probe.
for rollout_state, solver_time, solver_velocity in stored_old_trajectory:
    x_s = rollout_state
    s = solver_time
    u = solver_velocity
    compare_old_and_teacher(x_s, s, u)
```

3 Residual energies and the gate

Evaluate the frozen old and teacher predictors on the same z; both predict the native data-to-noise target u. Let D be the number of non-batch event coordinates and define

$$E_j(z,u) = \frac{1}{2c_D}\,\|u - f_j(z)\|_2^2, \qquad c_D = \begin{cases} 1, & \text{event sum,} \\ D, & \text{event mean.} \end{cases} \tag{4}$$

The teacher advantage and gate are

$$A_T = E_{\text{old}} - E_T, \qquad \gamma_T = \sigma\left(\operatorname{logit}\alpha + \frac{A_T}{\tau_s}\right), \qquad 0 < \alpha < 1, \quad \tau_s > 0. \tag{5}$$

Thus $A_T > 0$ means the teacher is closer to the sampled target than the old student. At $A_T = 0$, the gate is the prior α.

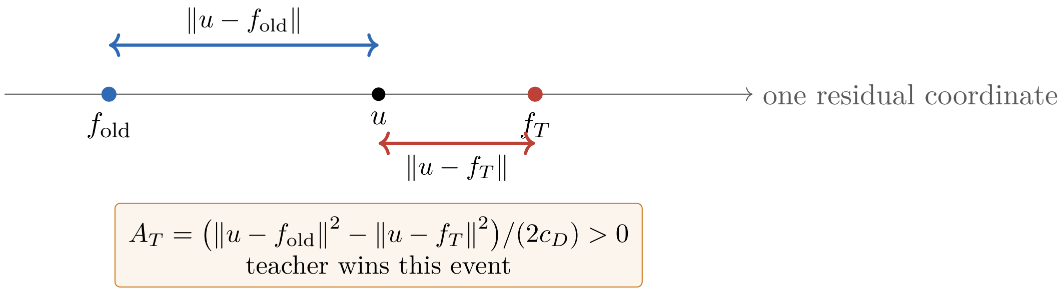


Figure 3: Residual-energy geometry in one coordinate. In D dimensions, squared Euclidean distances replace the intervals; the sign convention remains the same.

4 The exact auxiliary Gaussian interpretation

Proposition 4.1 (Exact posterior under shared predictive covariance)**.** *Fix z, assume $\sigma_r^2(s) > 0$, and introduce $B \in \{\text{old}, T\}$ with $\Pr(B = T) = \alpha$. Declare*

$$p_j(u \mid z) = (2\pi\sigma_r^2(s))^{-D/2} \exp\left[-\frac{\|u - f_j(z)\|^2}{2\sigma_r^2(s)}\right]. \tag{6}$$

Then

$$\Pr(B = T \mid u, z) = \sigma\left(\operatorname{logit}\alpha + \frac{c_D A_T}{\sigma_r^2(s)}\right). \tag{7}$$

Consequently, the gate in (5) is this exact posterior when

$$\boxed{\tau_s = \frac{\sigma_r^2(s)}{c_D}}. \tag{8}$$

Whenever A_T is nonzero on the relevant support, equality of the two logits there also requires this temperature.

Proof. Shared covariance makes the Gaussian normalizers cancel:

$$\log\frac{p_T(u \mid z)}{p_{\text{old}}(u \mid z)} = \frac{\|u - f_{\text{old}}\|^2 - \|u - f_T\|^2}{2\sigma_r^2(s)} = \frac{c_D A_T}{\sigma_r^2(s)}. \tag{9}$$

Posterior odds equal prior odds times the likelihood ratio. Taking logits gives (7); comparison with (5) gives (8). □

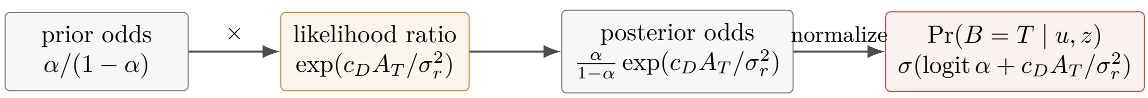


Figure 4: Bayes posterior and logit. The sign is positive because a lower teacher energy makes $A_T = E_{\text{old}} - E_T$ positive.

Declared residual covariance is not forward corruption noise. The covariance $\sigma_r^2(s)I_D$ in (6) models uncertainty of the *predictive label residual* $u - f_j(z)$ after conditioning on z. The fresh $\varepsilon \sim \mathcal{N}(0, I_D)$ in (2) creates the forward coupling and determines x_s. These are different random objects. There is no automatic identity such as $\sigma_r^2(s) = s^2$ or $\sigma_r^2(s) = 1$. Unequal old and teacher covariances would introduce log-determinant and unequal quadratic terms, so a plain energy difference would no longer be the exact posterior logit.

Exact likelihood versus calibrated surrogate

With event-sum energies and $\tau_s = \sigma_r^2(s)$, the gate is exact for the declared joint D-dimensional likelihood. With event-mean energies, exactness requires $\tau_s = \sigma_r^2(s)/D$. Choosing a resolution-stable numerical temperature, centering A_T, normalizing by a running scale, or clipping logits may be useful, but then the gate is a *calibrated surrogate*, not that exact joint-likelihood posterior.

5 What one sampled advantage estimates

Define the conditional mean and irreducible variance under old-generated endpoints and fresh forward noise:

$$m_{\mathrm{old}}(z) = \mathbb{E}[u \mid z], \qquad V_{\mathrm{old}}(z) = \mathbb{E}[\|u - m_{\mathrm{old}}(z)\|^2 \mid z]. \tag{10}$$

Proposition 5.1 (Conditional risk decomposition and cancellation)**.** *For either frozen predictor $j \in \{\mathrm{old}, T\}$,*

$$\mathbb{E}[E_j \mid z] = \frac{1}{2c_D}\left(V_{\mathrm{old}}(z) + \|f_j(z) - m_{\mathrm{old}}(z)\|^2\right), \tag{11}$$

and therefore

$$\mathbb{E}[A_T \mid z] = \frac{\|f_{\mathrm{old}} - m_{\mathrm{old}}\|^2 - \|f_T - m_{\mathrm{old}}\|^2}{2c_D}. \tag{12}$$

Thus A_T is a one-sample unbiased estimator of the conditional risk difference.

Proof. Write $u - f_j = (u - m_{\mathrm{old}}) + (m_{\mathrm{old}} - f_j)$ and expand. The conditional cross term vanishes because $\mathbb{E}[u - m_{\mathrm{old}} \mid z] = 0$. Subtracting the old and teacher expressions cancels the same irreducible variance V_{old}. □

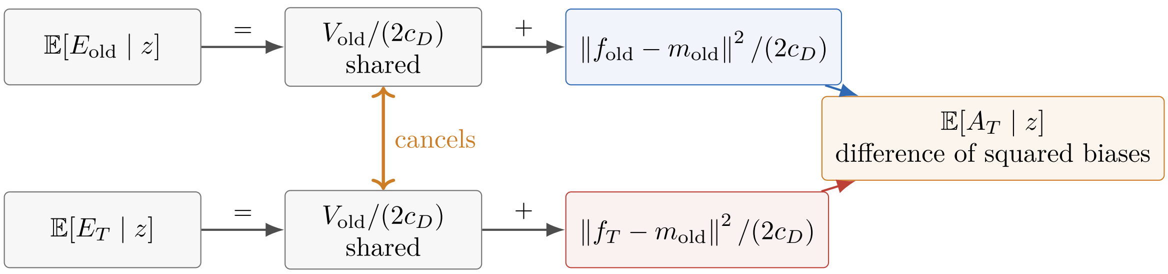


Figure 5: Conditional risk decomposition. A2 compares approximation biases after the shared irreducible label variance cancels.

Expectation and sigmoid do not commute. Although A_T is an unbiased estimator of the conditional mean risk difference, its transformed gate is generally not an unbiased estimator of the sigmoid of that mean:

$$\mathbb{E}[\gamma_T \mid z] \neq \sigma\left(\operatorname{logit}\alpha + \frac{\mathbb{E}[A_T \mid z]}{\tau_s}\right) \quad \text{in general.} \tag{13}$$

The discrepancy depends on the full conditional distribution of A_T, not only its mean.

Why the old field need not be the conditional mean. f_{old} generated x_{data} through an ODE, but $m_{\mathrm{old}}(z) = \mathbb{E}[u \mid x_s, s, C]$ belongs to a new coupling formed by re-noising the endpoint with fresh ε. Generation competence does not imply $f_{\mathrm{old}}(z) = m_{\mathrm{old}}(z)$ on that coupling. A2 only compares f_{old} and f_T under the old-endpoint probe distribution; it does not assume either is Bayes-optimal.

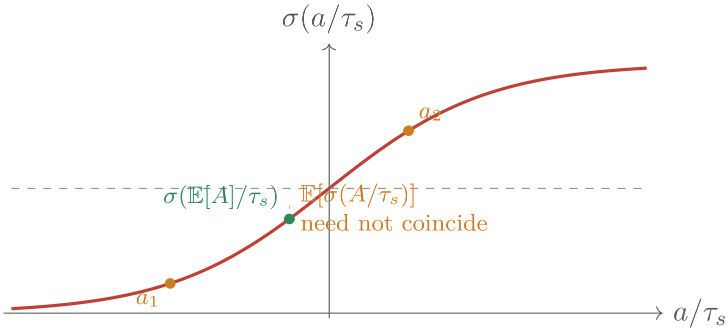


Figure 6: Noncommutation of expectation and sigmoid, schematically for a two-point advantage law. Equality occurs only in special cases (for example, a degenerate advantage).

6 Trust projection and the detached target

Let G_s be positive definite and define $\|a\|_{G_s}^2 = a^\top G_s a$. Let $r_s > 0$ be a G_s-norm output radius; it is an event-L^2 radius only when $G_s = I$. Set

$$d = f_T - f_{\text{old}}, \qquad c = \begin{cases} 1, & \|d\|_{G_s} = 0, \\ \min\{1, r_s / \|d\|_{G_s}\}, & \|d\|_{G_s} > 0. \end{cases} \tag{14}$$

The frozen, detached target and regression loss are

$$f_\gamma(z, u) = f_{\text{old}}(z) + \operatorname{sg}(\gamma_T) c\, d, \tag{15}$$

$$\mathcal{L}_{\text{A2}}(\theta) = \mathbb{E}\left[\frac{1}{2c_D} \|f_\theta(z) - f_\gamma(z, u)\|_{G_s}^2\right]. \tag{16}$$

The whole right-hand side of (15) is a target: old and teacher predictions are frozen, c is computed from frozen values, and γ_T is explicitly stop-gradient.

Proposition 6.1 (Trust bound and exact-replay gradient)**.** *For every event,*

$$\|f_\gamma - f_{\text{old}}\|_{G_s} \le r_s. \tag{17}$$

If $f_{\theta_{\text{old}}} = f_{\text{old}}$ exactly and $J_{\text{old}} = \partial f_\theta / \partial \theta|_{\theta_{\text{old}}}$, then

$$\nabla_\theta \mathcal{L}_{\text{A2}}|_{\theta_{\text{old}}} = -\frac{1}{c_D} \mathbb{E}\left[\gamma_T c\, J_{\text{old}}^\top G_s (f_T - f_{\text{old}})\right]. \tag{18}$$

Proof. Equation (15) gives $\|f_\gamma - f_{\text{old}}\|_{G_s} = \gamma_T c \|d\|_{G_s} \le c \|d\|_{G_s} \le r_s$. Differentiating (16) only through f_θ yields $c_D^{-1} J_\theta^\top G_s (f_\theta - f_\gamma)$. At replay, $f_\theta - f_\gamma = -\gamma_T c d$, proving the negative sign in (18). This is the aggregate regression gradient. For a single event, its self-Jacobian contribution is teacherward when the local Jacobian geometry permits; a shared minibatch parameter update also contains cross-event interactions and therefore does not guarantee pointwise teacherward movement for every sampled output. □

Listing 3: Good conceptual detached gate, projection, and student-only regression.

```
event_axes = tuple(range(1, u.ndim))
energy_old = (u - f_old).square().sum(dim=event_axes) / (2.0 * c_D)
energy_teacher = (
    (u - f_teacher).square().sum(dim=event_axes) / (2.0 * c_D)
)
advantage_teacher = energy_old - energy_teacher
gate = sigmoid(logit(alpha) + advantage_teacher / temperature_s)
```

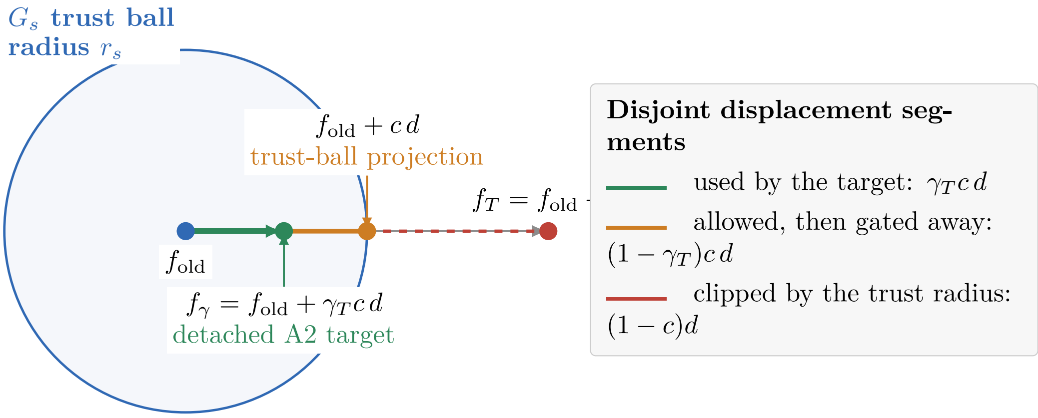


Figure 7: Trust-ball geometry. Projection acts first: $d \mapsto c\,d$ reaches at most the trust-ball boundary. The predictive-risk gate acts second: $c\,d \mapsto \gamma_T c\,d$. The three colored segments are disjoint, so labels do not obscure the distinction between accepted, gated-away, and radius-clipped displacement.

```

displacement = f_teacher - f_old
displacement_norm = metric_norm(displacement, metric_s)
projection = where(
    displacement_norm == 0,
    ones_like(displacement_norm),
    minimum(ones_like(displacement_norm), radius_s / displacement_norm),
)
target = (
    f_old + gate.detach() * projection * displacement
).detach()

f_student = student_model(x_s, s, condition)
loss_per_sample = metric_square(
    f_student - target, metric_s
).sum(dim=event_axes) / (2.0 * c_D)
loss = loss_per_sample.mean()
```

Listing 4: Bad: gradient leakage makes the gate and target gameable.

```
# WRONG: teacher/old/gate are part of the autograd graph.
f_old = trainable_student(x_s, s, condition)
f_teacher = teacher_model(x_s, s, condition)
gate = sigmoid(logit(alpha) + (energy(f_old) - energy(f_teacher)) / tau)
target = f_old + gate * (f_teacher - f_old)
loss = squared_error(trainable_student(x_s, s, condition), target)
loss.backward()  # Optimizes through the evidence and target construction.
```

RMS radius. If $G_s = I$ and the desired per-coordinate radius is r_s^{RMS}, then the corresponding event radius is $r_s = \sqrt{D}\, r_s^{\mathrm{RMS}}$. Mixing an RMS radius with an unnormalized event norm silently tightens the trust region as resolution grows.

7 Dimension, temperature, and saturation

For any $\delta \in (0, \frac{1}{2})$,

$$\gamma_T \le \delta \iff A_T \le \tau_s\big(\operatorname{logit}\delta - \operatorname{logit}\alpha\big), \tag{19}$$

$$\gamma_T \ge 1-\delta \iff A_T \ge \tau_s\big(\operatorname{logit}(1-\delta) - \operatorname{logit}\alpha\big). \tag{20}$$

At a fixed per-coordinate advantage a, event-sum evidence is approximately $A_T = Da$, whereas event-mean evidence is $A_T = a$. A fixed sum-gate temperature therefore becomes more binary as D grows.

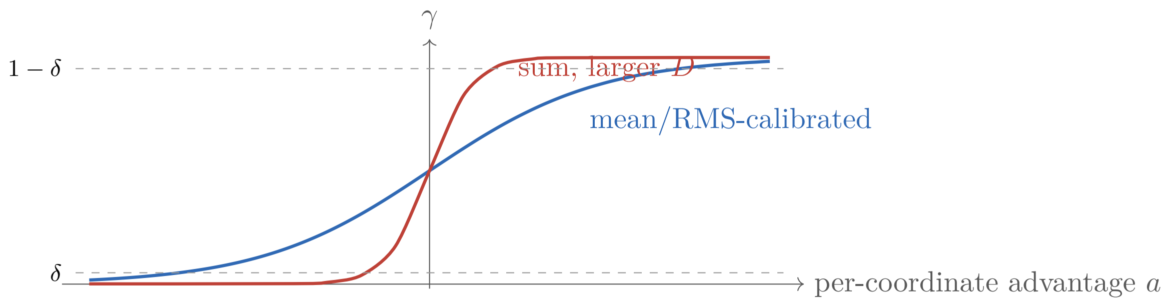


Figure 8: Event-sum versus event-mean saturation. The curves illustrate identical per-coordinate risk gaps with different effective logits; exact slopes depend on D/τ_s.

7.1 Resolution transfer rules

1. **Exact joint-likelihood convention.** Use event sums and $\tau_s = \sigma_r^2(s)$. If resolution changes, the log likelihood legitimately accumulates more coordinate evidence and the posterior may sharpen.

2. **Equivalent exact mean parameterization.** Divide energies by D and also divide temperature by D: $\tau_s = \sigma_r^2(s)/D$. The logit is unchanged.

3. **Practical resolution-stable surrogate.** Use event means and keep a numerical temperature on the per-coordinate scale. This deliberately changes the implied predictive covariance and must be labeled as calibration.

4. **Trust scale.** Transfer r_s^{RMS}, not a fixed event-L^2 radius, when D changes.

Listing 5: Good conceptual calibration diagnostics retain both event scales.

```
advantage_sum = energy_old_sum - energy_teacher_sum
advantage_per_dimension = advantage_sum / event_dimension

log_scalar("advantage_sum", advantage_sum)
log_scalar("advantage_per_dimension", advantage_per_dimension)
log_quantiles("gate", gate, probabilities=(0.01, 0.1, 0.5, 0.9, 0.99))
log_scalar("gate_entropy", binary_entropy(gate))
log_scalar("gate_low_saturation", mean(gate <= saturation_delta))
log_scalar("gate_high_saturation", mean(gate >= 1 - saturation_delta))
```

8 One inner step and stale behavior data

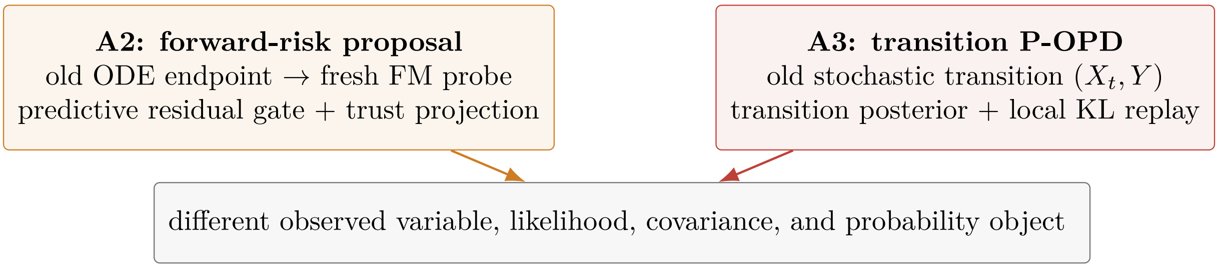
The cleanest A2 interpretation freezes an old model, collects behavior endpoints, constructs targets, and performs *one* inner optimizer step before refreshing the old snapshot and its data. At exact replay, Proposition 6.1 gives the intended local direction.

Several inner steps on the same batch are not mathematically forbidden, but after the first step $f_\theta \neq f_{\text{old}}$ while x_{data}, f_{old}, f_T, γ_T, c, and f_γ remain stale. Optimization then fits a fixed supervised target rather than repeatedly re-evaluating whether the teacher helps under the current behavior distribution. Large replay counts can move parameters far from the local trust interpretation even though every individual output target remains within radius r_s of the frozen old output. Output-space target trust is not a parameter-space or rollout-distribution trust guarantee.

Recommended semantic unit

Treat "freeze old $\to$ roll out $\to$ draw fresh probes $\to$ build detached targets $\to$ take one inner step $\to$ refresh" as the reference A2 unit. If replay is introduced for efficiency, ablate the replay count and measure realized post-update output displacement and endpoint drift.

9 A0–A4 and neighboring forward-probe methods



Shared surface form: $\sigma(\operatorname{logit}\alpha + \text{evidence})$.
This algebraic resemblance does not identify the methods.

Figure 9: A2 versus A3. The configured `p_opd` path is A3; A2 remains a distinct theory proposal based on fresh supervised forward-flow evidence.

arm	evidence distribution	target operation	mathematical status
A0	old-generated states	direct teacher regression	field matching under old occupancy
A1	old-generated states	fixed arithmetic interpolation or quadratic trust step	field-space geometry; no sample-dependent predictive test
A2	old ODE endpoints re-noised by fresh FM coupling	risk-gated, projected, detached teacher displacement	proposed supervised predictive-risk rule
A3	old SDE transition and sampled next state	transition-posterior gate with local replay	stochastic transition-mixture P-OPD; existing `p_opd`
A4	old/teacher source trajectory branch	regress sampled source velocity	implemented ODE time-marginal mixture through `marginal_cfm`

Relation to DPO, NFT, AWM, DGPO, and CRD. These decoupled methods share a useful mechanism: generate behavior outputs first, then sample fresh forward corruption times/noise and train on single-step forward probes. DPO uses preference pairs; DiffusionNFT forms a reward-weighted contrast between implicit positive and negative policies; AWM uses advantage-weighted matching; DGPO adds groupwise preference/advantage structure; and Centered Reward Distillation (CRD) matches centered external rewards to implicit rewards defined by forward prediction errors. This family resemblance motivates A2's solver-decoupled probe, but none of those objectives automatically supplies A2's Bayes interpretation, old-versus-teacher risk gate, or trust projection. Those must be derived and validated separately. The transition-posterior interpretation of A3 is related to DiffusionOPD and trust-region policy distillation [2, 3], while the distinct A4 marginal construction is developed in the internal A4 tutorial [6].

> **Terminology boundary**
>
> A2 is **not** a state-density-ratio method, **not** a transition-mixture KL method, and **not** a marginal-mixture method. Its exact probabilistic statement is only the auxiliary predictive likelihood $p_j(u \mid z)$ in Proposition 4.1.

10 Conceptual implementation blueprint

The following is an algorithmic blueprint, not a map to existing symbols.

1. Freeze an old snapshot and a teacher.
2. Solve the old ODE and retain its endpoint x_{data}.
3. Draw fresh s and ε; construct (x_s, u, z) using (3).
4. In no-gradient evaluation, compute $f_{\text{old}}(z)$ and $f_T(z)$ on the same input and in the same native target convention.
5. Compute A_T, a detached gate, and the metric projection using one declared sum/mean and temperature convention.
6. Detach f_γ completely.
7. Run exactly one gradient-enabled student forward on z and regress to f_γ.
8. After the update, measure displacement; refresh behavior data before interpreting another step as a fresh local A2 step.

Listing 6: Good high-level A2 blueprint with explicit invariant checks.

```python
# Conceptual pseudocode only; these names do not claim repository APIs.
x_data = old_ode_endpoint(condition, initial_noise).detach()
s, epsilon = fresh_forward_probe_randomness(x_data)
x_s, u = make_linear_forward_probe(x_data, s, epsilon)

with frozen_evaluation():
    f_old = old_predictor(x_s, s, condition)
    f_teacher = teacher_predictor(x_s, s, condition)

if f_old.shape != u.shape or f_teacher.shape != u.shape:
    raise ValueError(
        "expected old, teacher, and target shapes to match; "
        f"old={f_old.shape}, teacher={f_teacher.shape}, target={u.shape}"
```

```python
    )
if temperature_s <= 0:
    raise ValueError(
        f"expected positive A2 temperature, received {temperature_s!r}"
    )

gate, projection = detached_gate_and_projection(
    u=u,
    f_old=f_old,
    f_teacher=f_teacher,
    alpha=alpha,
    temperature_s=temperature_s,
    radius_s=radius_s,
    event_dimension=u[0].numel(),
)
target = (
    f_old + gate * projection * (f_teacher - f_old)
).detach()
f_student = student_predictor(x_s, s, condition)
student_forward_regression(f_student, target)
```

Listing 7: Bad: silent fallbacks erase the method's declared statistical meaning.

```python
# WRONG: do not hide invalid scales or missing predictions.
try:
    f_teacher = teacher_predictor(x_s, s, condition)
except Exception:
    f_teacher = f_old

temperature_s = max(temperature_s, 1e-8)
event_dimension = guessed_dimension if shape_is_confusing else x_s.numel()
gate = sigmoid((energy_old - energy_teacher) / temperature_s)
```

The bad listing is doubly dangerous: it swallows teacher failures and silently changes scale definitions. An implementation should validate types, shapes, finiteness, ranges, and target orientation before using them, then raise specific errors containing the received values and relevant identifiers.

11 Logging, calibration, and ablations

11.1 Minimum diagnostics

Log distributions, not only means:

- a consistently computed `advantage_sum` and `advantage_mean` = `advantage_sum`$/D$, including timestep-conditioned quantiles, plus an explicit record of which normalization drives the gate;
- γ_T quantiles, binary entropy $-\gamma \log \gamma - (1-\gamma)\log(1-\gamma)$, and low/high saturation rates;
- teacher-win rate $\Pr(A_T > 0)$;
- $\|d\|_{\mathrm{RMS}} = \|d\|_2 / \sqrt{D}$, $\|d\|_2$, and projection coefficient c;
- forward regression loss binned by s;
- realized post-update displacement $\|f_{\theta^+}(z) - f_{\mathrm{old}}(z)\|_{\mathrm{RMS}}$ and its ratio to the intended target movement;
- total and, where possible, componentwise gradient norms and clipping rates.

Gate statistics diagnose *whether* evidence selects the teacher; d and c diagnose *how large* the proposed movement is; realized displacement and gradient norms diagnose whether optimization actually executes that proposal.

11.2 Ablation matrix

axis	levels	hold fixed	decisive readout
energy normalization	sum; mean	same samples/models	separately labeled A_T^{sum} and A_T^{mean}, saturation versus D
temperature	exact σ_r^2/c_D; calibrated constant; s-schedule	normalization and prior	gate calibration, entropy, teacher-win-conditioned loss
prior α	low; 0.5; high	temperature	gate quantiles and update magnitude
trust radius	no clipping; RMS grid; $G_s = I$ event-L^2 grid	gate	c, realized displacement, stability
gate	constant α; hard win; sigmoid	same projected d	quality, variance, saturation sensitivity
probe time law	uniform; stratified; endpoint-biased	corruption formula	forward loss and mean-normalized advantage by s
inner replay	1; 2; 4+ steps per behavior batch	optimizer budget where feasible	stale-target drift and endpoint shift
resolution	matched low/high D	per-coordinate corruption	transfer of entropy, r^{RMS}, quality

For the exact-likelihood ablation, report the declared $\sigma_r^2(s)$ and verify (8). For calibrated surrogates, report the calibration transform explicitly and avoid posterior language unless empirical calibration is measured.

12 Limitations and recommended implementation order

12.1 Limitations

1. **Proposal status.** The derivation does not establish production correctness, distributed behavior, or gains on any model.

2. **Auxiliary model dependence.** Bayes exactness holds only for the declared shared covariance residual model and matched temperature.

3. **One-sample gate variance.** A_T is unbiased for a risk difference, but expectation and sigmoid do not commute; the gate can also be noisy or saturated.

4. **Old occupancy.** Evidence is gathered on endpoints produced by the old ODE; it says nothing directly about teacher-only regions.

5. **Conditional-mean mismatch.** Neither frozen model need equal $\mathbb{E}[u \mid z]$.

6. **Local trust only.** The output target is bounded at sampled probes, not globally in parameter space, endpoint distribution, or perceptual quality.

7. **Metric choice.** G_s, r_s, and the native prediction parameterization determine the geometry and require model-specific validation.

8. **Staleness.** Multi-step replay weakens the exact-replay local-gradient story.

9. **Compute.** Every probe requires frozen old and teacher predictions plus a gradient-enabled student prediction.

12.2 Recommended order

1. Build a CPU tensor-level reference for energies, posterior logits, projection, and the replay-gradient sign.
2. Verify prediction-target orientation and event axes on one real model batch.
3. Add diagnostic-only forward probes with no optimizer update; measure separately labeled sum- and mean-normalized advantages, gate saturation, d scales, and timestep dependence.
4. Choose either exact joint-likelihood scaling or an explicitly named calibrated surrogate.
5. Add detached targets and one inner step with a conservative RMS trust radius.
6. Measure realized displacement, gradients, and endpoint drift before increasing replay or radius.
7. Run the normalization, temperature, radius, and resolution ablations in Section 11.
8. Only after single-device numerical validation, design distributed call order, snapshot lifecycle, checkpointing, and production configuration.

13 Summary

Arm A2 separates three questions:

$$\boxed{\begin{array}{rl} \text{Where is evidence measured?} & x_{\text{data}} \xrightarrow[\varepsilon,s\ \text{fresh}]{\text{forward FM}} (z,u), \\ \text{Does the teacher help?} & A_T = E_{\text{old}} - E_T, \quad \gamma_T = \sigma(\operatorname{logit}\alpha + A_T/\tau_s), \\ \text{How far may the target move?} & c = \min(1, r_s/\left\|d\right\|_{G_s}), \quad f_\gamma = f_{\text{old}} + \operatorname{sg}(\gamma_T)cd. \end{array}}$$

The shared-covariance posterior is obtained with $\tau_s = \sigma_r^2(s)/c_D$, and this scale is identifiable wherever the advantage is nonzero. The risk advantage is one-sample unbiased because irreducible conditional variance cancels, but expectation and sigmoid generally do not commute. The detached target yields the negative replay-gradient sign in (18); its single-event self-Jacobian contribution is teacherward when the local geometry permits, without a pointwise minibatch guarantee. These statements define a forward-risk proposal; they do not turn A2 into A3, A4, or an existing target mode.

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